

Module 2 Reflection Journal

In this lab, I learned more about the main tools used in machine learning. Before this lab, I knew a little about Python, but I was not very familiar with how different libraries work together. After completing the exercises, I understood better why tools like Pandas, NumPy, Matplotlib, Scikit-learn, Google Colab, and GitHub are important for machine learning.

One thing I learned was how Pandas can make data easier to understand. We used the Iris dataset and converted the data into a DataFrame. Seeing the data in rows and columns helped me understand the information more clearly. I also learned how to check the first few rows of the dataset and view information about the columns. This helped me see the data types and understand how the dataset was organized.

I also learned more about NumPy. In the assessment, I used NumPy to calculate the mean and standard deviation of sepal length. At first, these calculations seemed like simple math, but I realized that NumPy can make these operations much easier when working with a large amount of data. Instead of calculating values manually, we can use built-in functions and get the results quickly.

Matplotlib was another useful tool that I practiced in this lab. I created a scatter plot using sepal length and sepal width. The graph helped me see that the three Iris species have different patterns, although some of the data points overlap. I also created a bar chart to show how many samples there were for each species. This helped me understand that visualizations can make data easier to analyze compared to only looking at numbers.

One challenge for me was understanding how all the tools connect together. At first, Pandas, NumPy, Matplotlib, and Scikit-learn seemed like separate tools. After working through the lab, I started to understand that they can be used together in one machine learning workflow. Scikit-learn can provide the dataset, Pandas can organize the data, NumPy can perform calculations, and Matplotlib can show the results visually.

I also learned that checking the output after running each code cell is very important. If there is an error, reading the error message and checking the code can help find the problem. Running the notebook step by step also helped me understand what each section of the code was doing instead of just running everything without looking at the results.

Another thing I learned from this lab is the importance of GitHub. I understood that GitHub is not only a place to upload files. It can also be used to organize projects, keep track of work, and share files with team members. This will be useful for future machine learning projects because there may be many notebooks and project files that need to be organized.

Overall, this lab helped me feel more comfortable using machine learning tools. I learned how to load and explore a dataset, organize data, perform simple calculations, and create visualizations. I still need more practice, but I now understand the basic workflow better and feel more prepared for future machine learning labs.

